MASTER FUNCTIONAL SKILLS (MFS) · WORKED SOLUTIONS ·
MasterFunctionalSkills.co.uk · AQA Maths Level 1 Specialist

Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

Surname _____

Forename(s) _____

Candidate signature _____

Functional Skills Certificate
FUNCTIONAL MATHEMATICS

Level 1

Monday 14 January 2019 Morning Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:
- a calculator
- mathematical instruments
- a copy of the Data Book (Examination) (enclosed).



For Examiner's Use	
Question	Mark
1	
2	
3	
4	
TOTAL	

Instructions
- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- State the units of your answer where appropriate

Inform
- The
- The
- You
  The
- Evid
  ques

Advice
- In al

OFFICIAL MFS MODEL SOLUTION · BENCHMARK EXEMPLARY PASS

Full Marks / Exemplary (100% Pass · Model Solutions)

· Assessment: January 2019

· Subject: AQA Functional Skills Maths Level 1

· Complete accredited step-by-step models and tutor guidance

· Exam Board: AQA (Ofqual Regulated Reform Standard)

Verified by Tutor Abdul (Senior Assessor) · MasterFunctionalSkills.co.uk

MASTER FUNCTIONAL SKILLS

ONLINE LEARNING PLATFORM & ACCREDITED COURSES

Pearson Edexcel & Open Awards Accredited Suite · Entry 3, Level 1 & Level 2

Pass Your Functional Skills Exams With Confidence

100% Pass Guarantee | Dyslexia-Friendly Learning | 1-to-1 Expert Tutor Support

Master Functional Skills provides an adaptive, dyslexia-friendly online learning platform tailored for adult learners, apprentices, and NHS healthcare staff. Everything you need to revise, practice, and pass is at your fingertips.

1. Smart Diagnostic Initial Assessment

Calculate your exact working level and pinpoint knowledge gaps instantly

- **Exact Grade Calculation:** Adaptive assessments calculate your exact level across all topics.
- **Personalised Study Pathway:** Skip what you know; focus 100% on the topics needed to pass.
- **Real-Time Exam Readiness:** Predictive tracker confirms when you are 100% ready to book.

2. Comprehensive Video Lessons & Worked Solutions

Bite-sized visual explainers covering every single Maths & English curriculum standard

- **Step-by-Step Video Lessons:** Master percentages, algebra, perimeter, ratio, and reverse fractions.
- **Multi-Ink Handwriting Models:** Visual 3-colour methods showing working, carries, and examiner tips.
- **Tutor Shortcuts & Golden Rules:** Proven non-calculator mental arithmetic hacks and BIDMAS tricks.

3. Full Mock Exams & Real-Time Performance Analytics

Practice in exam conditions and watch your scores and confidence climb

- **Authentic Exam Papers:** Timed Section A & Section B papers formatted exactly like Pearson Edexcel.
- **Instant Topic Breakdown:** View percentage scores across Number, Measure/Shape, and Data.
- **Historical Score Tracking:** Monitor past attempts to ensure continuous improvement before exam day.

Ready to Master Your Functional Skills & Pass First Time?

Join thousands of healthcare staff, apprentices, and adult learners across the UK.

Instant 24/7 access on desktop, tablet, and mobile with no software to install.

START LEARNING TODAY AT

MasterFunctionalSkills.co.uk

Answer **all** questions in the spaces provided.

outside the box

1 **Part-time work**

There is a **data sheet** for Part-time work.

Emmie is a student.

She lives near Queens Road.



I work one day each week in a shop in the Trafford Centre.

Emmie

Here are the details of Emmie's journey to work.

5-minute walk from home to the Queens Road tram stop
Get the tram from Queens Road to Piccadilly Gardens
Get the bus from Piccadilly Gardens to the Trafford Centre
10-minute walk from the bus stop at the Trafford Centre to the shop

outside the box

Emmie starts work at 10.00 am

She leaves home at 8.30 am to catch the tram at Queens Road.

The tram is on time.

1 (a) What time does she get to Piccadilly Gardens?

[2 marks]

8:30 + 5 min = 8:35 arrive at tram stop.

Get on tram at 8:37

⇒ Arrive at PG at 8:48 am.

1 (b) Emmie says,

"I should arrive at the shop before 10.00 am"

Is she correct?

You **must** show your working.

[5 marks]

Next bus from PG to TC is at 9:12am.

Arrives at TC at 9:45am.

9:45am + 10 min = 9:55am.

Yes, she is correct.

outside the box

1 (c) One day, Emmie is paid £41.30

She pays these fares.

Queens Road to Piccadilly Gardens	£4.00 return
Piccadilly Gardens to the Trafford Centre	£3.70 return

Emmie says,

"After paying for fares I have **more than** £35 of my pay left over."

Is she correct?
You **must** show your working.

[4 marks]

£4 + £3.70 = £7.70

£41.30 - £7.70 = £33.60

No, she is incorrect.

†1 (d) Gail works with Emmie.

She is 21 years old.

She is paid the National Minimum Wage.

She works for 8 hours.

How much is she paid?

[2 marks]

£7.38 × 8 = £59.04

Check your answer.
Show how you have done your check.

[1 mark]

£59.04 ÷ £7.38 = 8 hours worked.

14

outside the box

2 **On the farm**

There is a **data sheet** for On the farm.



I am a farmer.
I keep cows and pigs.

Ken

2 (a) One day, Ken records the milk yields, in litres, for 10 of his cows.

20.82	22.03	22.24	21.42	21.36
21.91	22.14	23.12	21.81	21.55

The UK mean milk yield per cow is 21.76 litres per day.

Ken says,

"The mean milk yield for these 10 cows today is **more than** 21.76 litres."

Is he correct?
You **must** show your working.

[4 marks]

$20.82 + 22.03 + \ldots + 21.81 + 21.55 = 218.4\text{L}$ total.

$\frac{218.4}{10} = 21.84\text{L}$

Yes, he is correct.

Question 2 continues on the next page

2 (b) In total, Ken has 150 cows.

The mean milk yield per cow is 7944 litres per year.

Ken sells all the milk for 33p per litre.

This year, his total costs for feeding and looking after the cows are £333 600

He says,

"This year, my profit on the milk from these cows is **more than** £60 000"

Is he correct?

You **must** show your working.

[5 marks]

box

7944 × 150 × £0.33 = £393228.

£393228 − £333600 = £59628

No, he is incorrect.

outside the box

†2 (c) Ken wants to build a sty for one sow and one piglet.
Work out the minimum area of sty he needs to build.

[2 marks]

2.48 + 1.8 = 4.28 m²

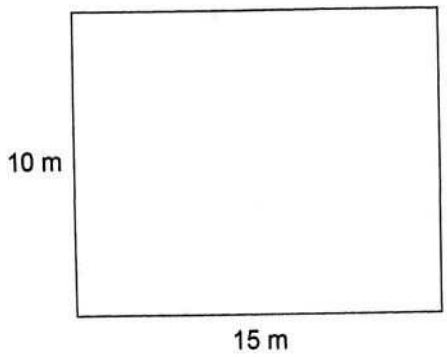
Check your answer.
Show how you have done your check.

[1 mark]

4.28 − 1.8 = 2.48m².

Question 2 continues on the next page

2 (d) Here is a sketch of the fence around some land that Ken will use for new sties.



Not drawn accurately

Ken wants five square sties around the edges of the land, each 4 m by 4 m

The rest of the land will be open space.

Each sty must have at least one side facing the open space.

In the open space he wants

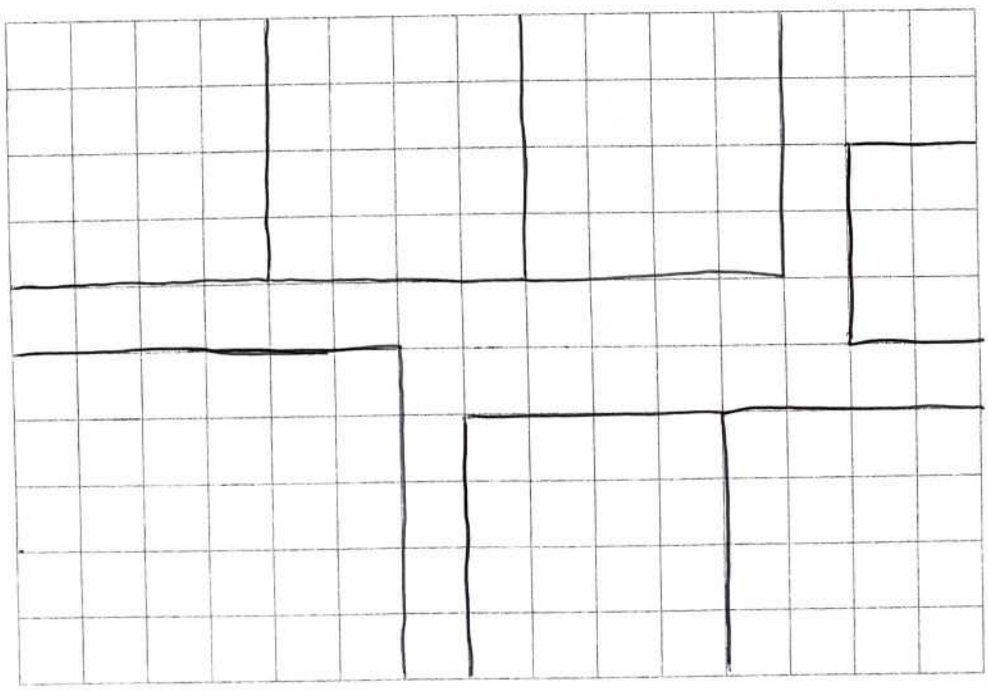
- a rectangular digging area of at least 30 m^2
- a rectangular feeding trough, 2 m by 3 m

Show a possible design for the land.

[3 marks]

Practise on this grid.

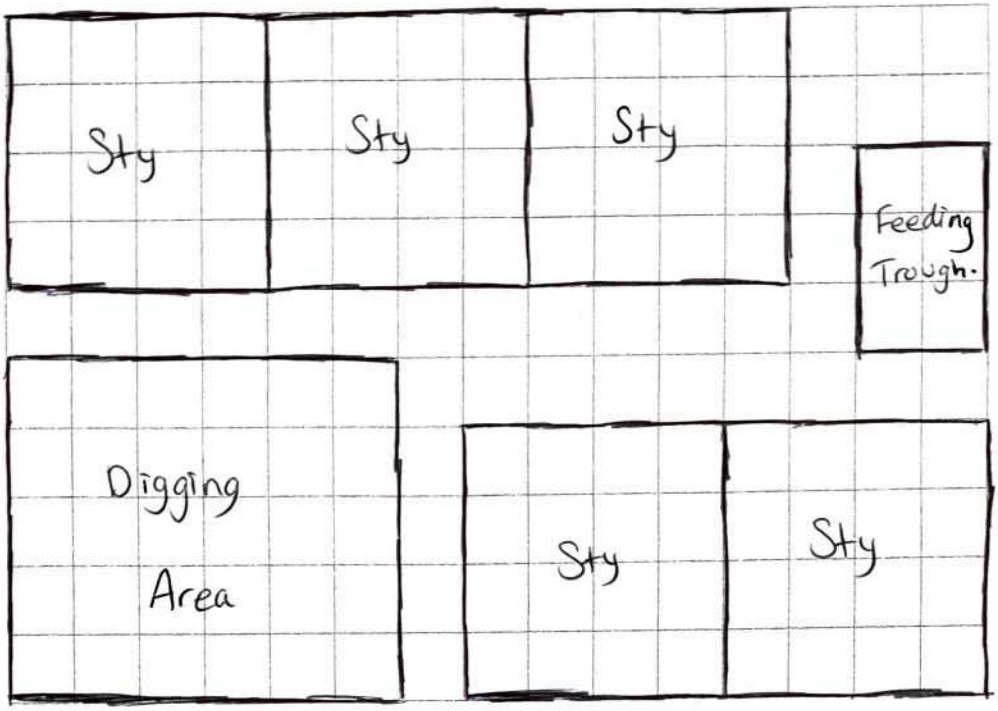
Scale 1 cm represents 1 m



outside the box

Put your answer on this grid.

Scale 1 cm represents 1 m



15

Turn over for the next question

3 **Log burner**

box

Rohan wants to put a log burner in his living room.



Here is some information about three log burners.

The log burners are called the Brunel, the Chester and the Dover.

		Dimensions of log burner (mm)		
	Heat output (kW)	**Height**	**Width**	**Depth**
Brunel	4	524	385	290
Chester	6	592	553	397
Dover	11	678	759	403

3 (a) Rohan says,

"The width of the Dover is **more than** double the width of the Brunel."

Is he correct?

You **must** show your working.

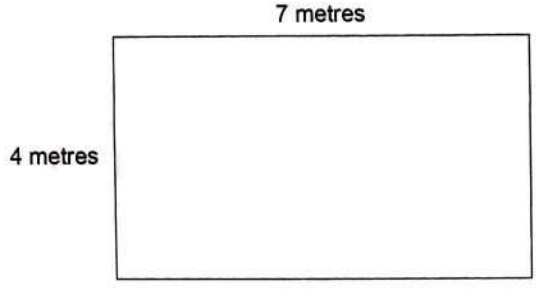
[3 marks]

$\frac{759}{385} = 1.971$

No, he is not correct.

3 (b) The height of Rohan's living room is 3 metres.

Here is a floor plan of the room.



Not drawn accurately

Rohan uses these steps to work out the heat output needed for the room.

Step 1	Work out	length of room × width of room
Step 2	Work out	answer to **Step 1** × height of room
Step 3	Work out	answer to **Step 2** ÷ 14

What is the smallest log burner with enough heat output for this room?
You **must** show your working.

[5 marks]

$7 \times 4 = 28\,m^2$.

$28\,m^2 \times 3m = 84\,m^3$

$\frac{84}{14} = 6\,kW$ required.

⇒ Chester.

Question 3 continues on the next page

3 (c) Dania has a log burner.

The log burner can burn smokeless fuel or wood.

She finds these costs.

- 16 kg bag of smokeless fuel costs £7.50
- 12 kg bag of wood costs £6.90

Dania's log burner uses 2 kg of smokeless fuel or wood per hour.

She uses the log burner for 24 hours per **week.**

Dania says,

"Using smokeless fuel is cheaper than using wood by **more than** £5 per week."

Is she correct?

You **must** show your working.

[7 marks]

box

$24 \times 2kg = 48\ kg$ / week.

SF: $\frac{48}{16} = 3$ bags

$3 \times £7.50 = £22.50$

Wood: $\frac{48}{12} = 4$ bags

$4 \times £6.90 = £27.60$.

$£27.60 - £22.50 = £5.10$

Yes, she is correct.

outside the box

15

Turn over for the next question

4 **Apricot jam**

Here is a recipe for apricot jam.

Ingredients for one batch of jam	
Apricots	2 kg
Sugar	600 g
Water	250 ml

Method

Put the ingredients into a pan

Heat the mixture to 105 °C

Cook for 40 minutes

4 (a) Convert 250 ml to litres.

Circle your answer.

[1 mark]

0.025 litres 0.25 litres 2.5 litres 25 litres

4 (b) Peter is going to use this recipe to make apricot jam.

He has

8 kg of apricots

2500 g of sugar.

He says,

"I have enough sugar to make the jam using **all** 8 kg of apricots."

Is he correct?

You **must** show your working.

[4 marks]

$\frac{8}{2} = 4$ batches.

4 × 600g = 2400g (sugar) needed.

Yes, he is correct.

Peter starts to heat the mixture.

4 (c) This thermometer shows the temperature of the mixture after a few minutes.



How many degrees hotter does the mixture need to be to reach 105 °C?

[2 marks]

105 − 98·4 = 6·6 °C.

4 (d) The temperature reaches 105 °C at 3.35 pm

At what time should Peter finish cooking the jam?

[2 marks]

3:35 + 40 mins = 4:15 pm.

4 (e) Peter makes 20 pounds of jam.

1 pound = 16 ounces

He puts the jam into jars.

Each jar holds 10 ounces of jam.

How many jars can Peter fill with jam?

[3 marks]

20 lbs × 16 oz/lb = 320 oz.

$\frac{320}{10}$ = 32 jars.

4 (f) Sandra also makes apricot jam.

She makes 25 jars of jam.

Each jar of jam costs £3.04 to make.

She sells each jar of jam for £3.99

She says,

"I have made a profit of **more than** £24"

Is she correct?

You **must** show your working.

[4 marks]

box

£3.99 - £3.04 = £0.95 profit per jar.

£0.95 x 25 = £23.75

No, she is incorrect.

16

END OF QUESTIONS

box

There are no questions printed on this page

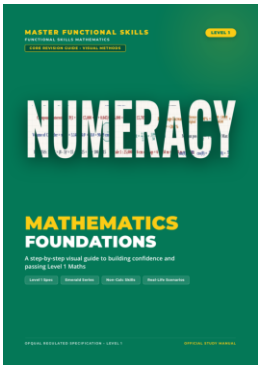
DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED

MASTER FUNCTIONAL SKILLS STORE

OFFICIAL REVISION GUIDES, WORKBOOKS & FLASHCARDS

Written by accredited specialist tutors to match the exact 2026 Pearson Edexcel standards

Everything You Need To Revise, Practice & Pass Your Maths Level 1 Exam



Core Curriculum Manual

Maths Level 1
Core Study Guide

- Full official curriculum coverage
- Section A Non-Calc & Sec B Calc
- Worked models & golden rules
- Dyslexia-friendly visual format

Spiral Bound Print & PDF

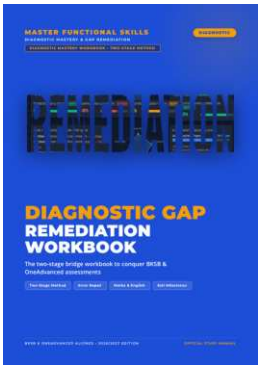


1,000+ Practice Drills

Maths Level 1
Companion Workbook

- 1,000+ exam-style questions
- Progressive difficulty drills
- Full worked step-by-step key
- Targeted fix for weak areas

Spiral Bound Print & PDF



Realistic Mock Papers

Maths Level 1
Official Mock Packs

- Full mock exam paper sets
- Authentic Pearson Edexcel style
- Detailed mark schemes included
- Grade boundaries & pass marks

Exam Booklet Pack & PDF



Pocket Flashcards A6

Maths Level 1
Pocket Revision Cards

- Compact A6 cards for quick study
- Essential formulas & conversions
- Dual-sided question & answer
- High-yield mental arithmetic rules

Boxed Card Deck & Print PDF

Order Official Study Guides & Revision Cards Online

Free Next-Day UK Tracked Shipping · Instant PDF Downloads Included

Trusted by NHS trusts, apprentices & adult learners nationwide.

VISIT THE ONLINE STORE

MasterFunctionalSkills.co.uk